S. B. JAIN INSTITUTE OF TECHNOLOGY, MANAGEMENT & RESEARCH, NAGPUR.

Practical No. 8

Aim: Demonstrate the concept of Semantic network on imputed predicate logic to represent the knowledge.

Name of Student: _____

Roll No.: _____

Semester/Year: **VI/III**

Academic Session: **2025-2026**

Date of Performance: _____

Date of Submission: _____

AIM: Demonstrate the concept of Semantic network on imputed predicate logic to represent the knowledge.

OBJECTIVE/EXPECTED LEARNING OUTCOME:

The objectives and expected learning outcome of this practical are:

- Understand semantic networks as a knowledge representation technique
- Implement ISA and HAS-A relationships
- Demonstrate inheritance in semantic networks
- Query knowledge programmatically using Python

THEORY:

Semantic Network

A **semantic network** is a graphical knowledge representation technique consisting of:

- **Nodes** → Concepts / Objects
- **Edges** → Relationships

Common relations:

- ISA → Inheritance
- HAS-A → Property
- CAN → Ability

It is closely related to **predicate logic representation**, where:

isa(dog, animal).
has(dog, tail).
can(bird, fly).

Relationship to Predicate Logic

A semantic network can be represented using first-order predicates:

Semantic Relation	Predicate Form
Dog is an Animal	isa(dog, animal)
Bird can Fly	can(bird, fly)
Dog has Tail	has(dog, tail)

Inheritance rule:

If isa(X, Y) and has(Y,

Z)

Then

has(X, Z)

Semantic Network Example

Nodes:

- Animal
- Bird
- Dog
- Sparrow

Relationships:

- Bird ISA Animal Dog ISA Animal
- Sparrow ISA Bird
- Animal has cells
- Bird can fly
- Dog can bark

SEMENTIC NETWORK DIAGRAM:

PROGRAM CODE:

```
class SemanticNetwork:
    def __init__(self):
        self.relations = []
    def add_relation(self, subject, predicate, obj):
        self.relations.append((subject, predicate, obj))
    def show_relations(self):
        print("Knowledge Base:")
        for s, p, o in self.relations:
            print(f"{p}({s}, {o})")
    def infer(self, entity):
        results = []
        stack = [entity]
        while stack:
            current = stack.pop()
            for s, p, o in self.relations:
                if s == current and p == "is_a":
                    stack.append(o)
                    results.append((current, "is_a", o))
                if s == current and p == "can":
                    results.append((current, "can", o))
        return results
sn = SemanticNetwork()
sn.add_relation("Canary", "is_a", "Bird")
sn.add_relation("Bird", "is_a", "Animal")
sn.add_relation("Bird", "can", "Fly")
sn.add_relation("Tweety", "is_a", "Canary")
sn.show_relations()
print("\nInference for Tweety:")
inferred = sn.infer("Tweety")
for rel in inferred:
    print(f"{rel[1]}({rel[0]}, {rel[2]}
```

INPUT & OUTPUT:

P8.py

```python
class SemanticNetwork:  1 usage
    def __init__(self):
        self.relations = []
    def add_relation(self, subject, predicate, obj):  4 usages
        self.relations.append((subject, predicate, obj))
    def show_relations(self):  1 usage
        print("Knowledge Base:")
        for s, p, o in self.relations:
            print(f"{p}({s}, {o})")
    def infer(self, entity):  1 usage
        results = []
        stack = [entity]
        while stack:
            current = stack.pop()
            for s, p, o in self.relations:
                if s == current and p == "is_a":
                    stack.append(o)
                    results.append((current, "is_a", o))
                if s == current and p == "can":
                    results.append((current, "can", o))
        return results
sn = SemanticNetwork()
sn.add_relation("Canary", "is_a", "Bird")
sn.add_relation("Bird", "is_a", "Animal")
sn.add_relation("Bird", "can", "Fly")
sn.add_relation("Tweety", "is_a", "Canary")
sn.show_relations()
print("\nInference for Tweety:")
```

```
C:\Users\LAB-112\AppData\Local\Programs\Python\Python313\python.exe
Knowledge Base:
is_a(Canary, Bird)
is_a(Bird, Animal)
can(Bird, Fly)
is_a(Tweety, Canary)

Inference for Tweety:
is_a(Tweety, Canary)
is_a(Canary, Bird)
is_a(Bird, Animal)
can(Bird, Fly)
```

CONCLUSION: **Hence we understood the concept of Semantic Network also we Demonstrate the concept of Semantic network on imputed predicate logic to represent the knowledge.**

DISCUSSION QUESTIONS:

1. What is a semantic network?

_____

_____

_____

2. What is ISA relationship?

_____

_____

_____

3. What is inheritance in knowledge representation?

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_____

_____

4. How is predicate logic used here?

_____

_____

_____

5. What are advantages of semantic networks?

_____

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_____

REFERENCES:

- *https://www.geeksforgeeks.org/artificial-intelligence/semantic-networks-in-artificialintelligence/*
- *https://thecloudflare.com/knowledge-representation-using-the-semantic-network-in-ai/*
- *https://www.siliconvalleycloudit.com/knowledge-representation-in-ai-semantic-networks/*